

VARUNKUMAR SONAWANE

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Summary

AI-focused Software Development Engineer with hands-on experience building **scalable backend systems** and data pipelines, strong foundations in data structures and problem solving, and experience developing production-ready applications and contributing to **open-source projects**, integrating **AI agents and LLM-based automation** on **AWS** and **GCP** using **Python** and **SQL**.

Education

Indiana University Bloomington, IN, USA.

Master of Science in Computer Science

Aug 2024 – May 2026

CGPA: 3.83/4

Pune University, MH, INDIA.

Bachelor's degree in Computer Science

Aug 2019 – May 2023

CGPA: 9/10

Certifications

[AWS Certified Developer – Associate](#) | [Microsoft Azure AI Fundamentals](#)

Experience

Graduate Teaching Assistant, Indiana University, Bloomington, IN

Aug 2025 - Present

- Mentored **120+** students on core software engineering topics, providing detailed feedback on **code quality, documentation, and object-oriented design**.
- Guided teams through end-to-end development processes including design, implementation, testing, deployment, and debugging of distributed systems, NLP pipelines, and web scraping applications using Python, Scala, and Spark.
- Collaborated** with the instructor to develop assignments, project rubrics and coding labs on **SQL, version control, Agile workflows, and Tableau dashboards**, strengthening end-to-end engineering practices.

Data Engineer Intern, The Commons XR, San Diego, CA

May 2025 - Aug 2025

- Designed and implemented scalable backend systems and data pipelines processing **5M+** records per week, improving system reliability and throughput from PostgreSQL to BigQuery, eliminating manual handoffs and improving system reliability.
- Integrated an **LLM-powered automation** service on Vertex AI to replace manual post-session analysis, generating structured insights and reducing recurring engineering effort.
- Created and maintained **10+** production features and services in **Python, Plotly, and Dash** to support real-time and **ad-hoc** analytics for XR platforms.

Software Engineering Intern, CodeClause, India

July 2023 - Jan 2024

- Built a cloud-based pipeline on **AWS (S3, Lambda, DynamoDB, ECS Fargate)** using Docker and CI/CD workflows, supporting data processing and troubleshooting issues across services during the software development lifecycle (**SDLC**).
- Developed microservices using Python with a React frontend, processing **10K+** daily records while participating in code reviews and debugging distributed systems in an agile environment using Git.

Projects

VoiceLegal-AI Web Extension

- Built a **conversational AI** system for legal document analysis integrating **web client, backend services, and Chrome extension**, achieving **4.8×** faster responses by optimizing service **latency** and **backend workflow**.
- Elevated user experience** by integrating **Vertex AI (Gemini 2.5)** and **ElevenLabs voice**, delivering explainable natural-language responses through chat and voice.
- Designed **fault tolerant, high availability** distributed system supporting 200+ users through **scalable service architecture**.

IdeaGenie - AI-Driven Idea Evaluation Platform | *Luddy Hackathon Winner*

- Developed a **full-stack AI** product by developing a Flask REST API and a React frontend to evaluate and rank top 3 user ideas, enabling automated prioritization and reducing manual review effort.
- Improved decision quality and usability** by integrating LLM-based reasoning with an interactive UI, allowing users to adjust parameters and receive explainable, real-time feedback.

DocWeaver: Healthcare Document Platform | *Python, TypeScript, FastAPI, Next.js, Gemini AI*

- Orchestrated AI platform automating medical documentation through **10+ specialized agents** that process clinical documents and generate patient reports, reducing documentation time by 80%.
- Implemented async processing with rate limiting using Python and FastAPI to handle distributed workflows efficiently across multiple AI agents.

Technical Skills

Languages: Python, SQL, HTML5, CSS3, JavaScript

Cloud & Databases: AWS (EC2, S3, Lambda, Glue, RDS), GCP (Vertex AI, BigQuery, GCS), PostgreSQL, MySQL, MongoDB

Backend & Systems: RESTful APIs, Distributed Systems, API Versioning, Load Balancing, Linux

Distributed & Data: Spark, Airflow, Query Optimization

AI/ML & Tools: TensorFlow, Scikit-learn, LangChain, Git, Jenkins, Kubernetes